ENG3000

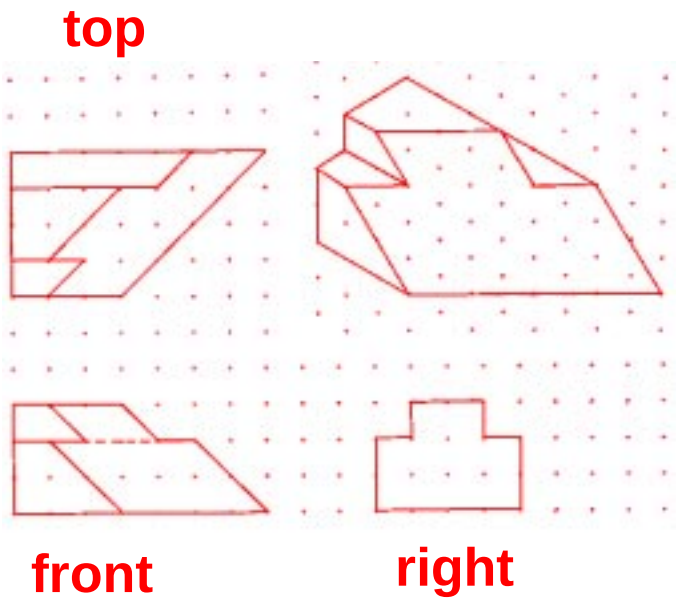
Orthographic Projection

Session Objectives

- To understand the basic principles of orthographic projection
- To be able to construct orthographic views of simple objects
- To visualize 3−D objects from drawings showing their orthographic views

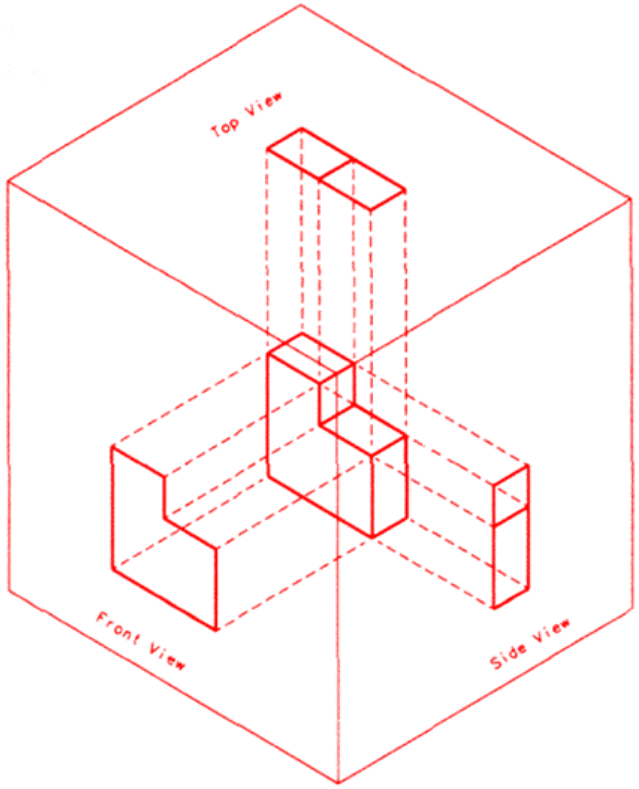
Orthographic Projection

- *Ortho* – Greek word meaning perpendicular
- Show the views of an object projected in 2–D
- Usually the top, front, and right side views



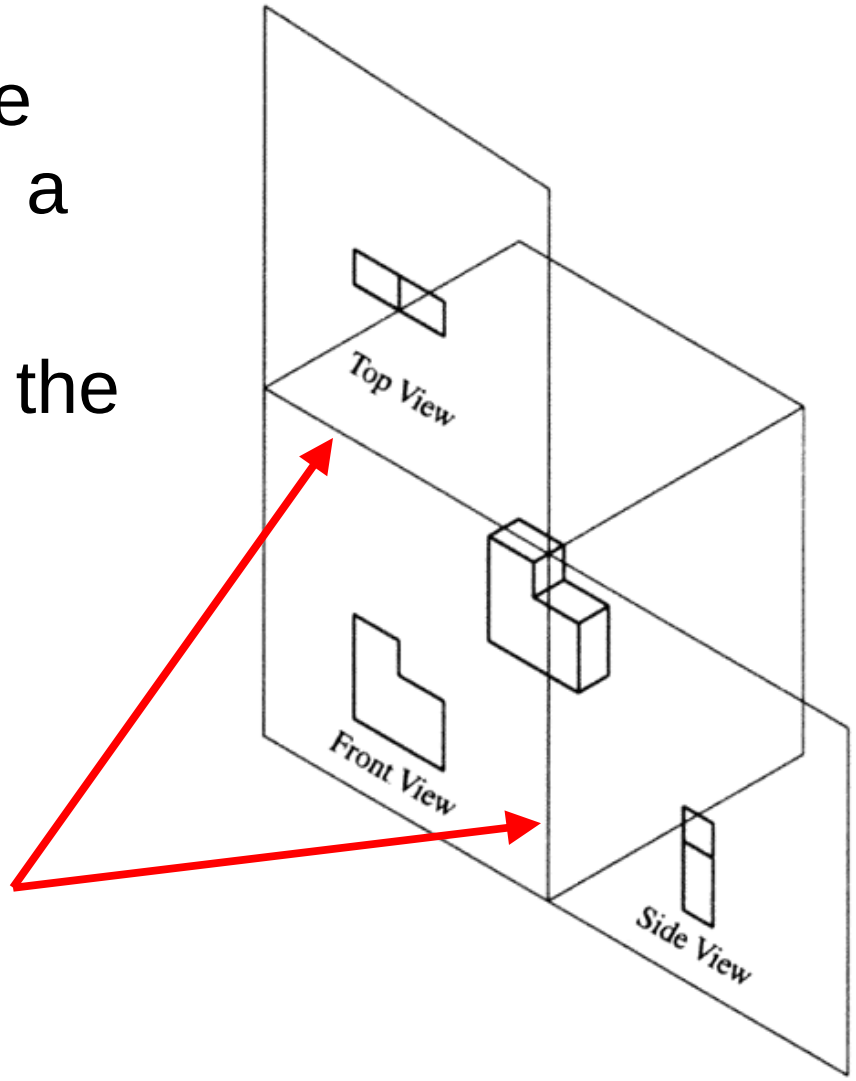
Orthographic Projection

- Imagine the object is surrounded by a glass cube.
- Object surfaces are projected onto the faces.



Orthographic Projection

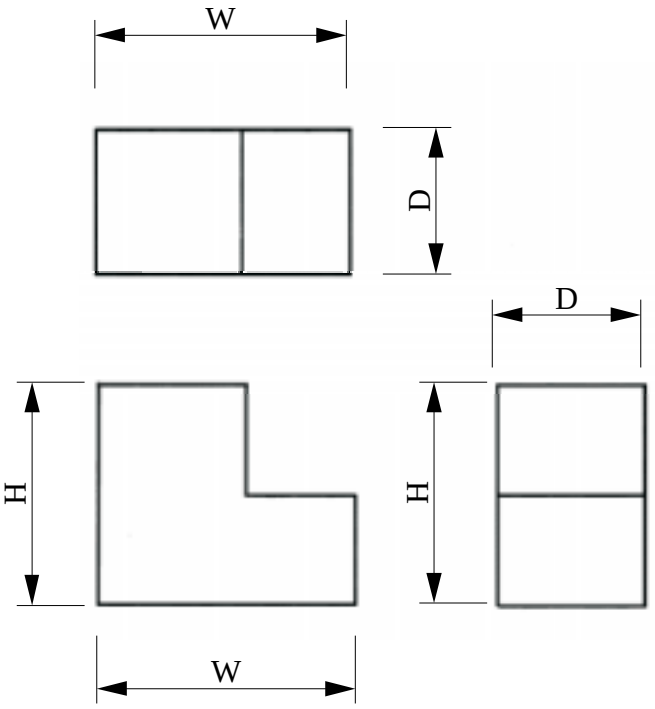
- Unfold the cube so that it lies in a single plane
- Three views of the object are now visible on the same plane in space



Orthographic Projection

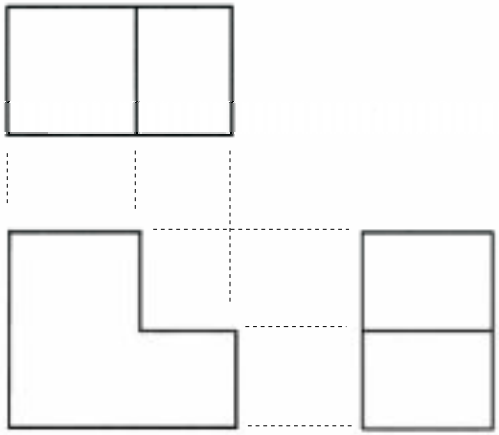
When the glass cube is unfolded:

- Front view: Height and Width
- Top view: Width and Depth
- Right view: Depth and Height

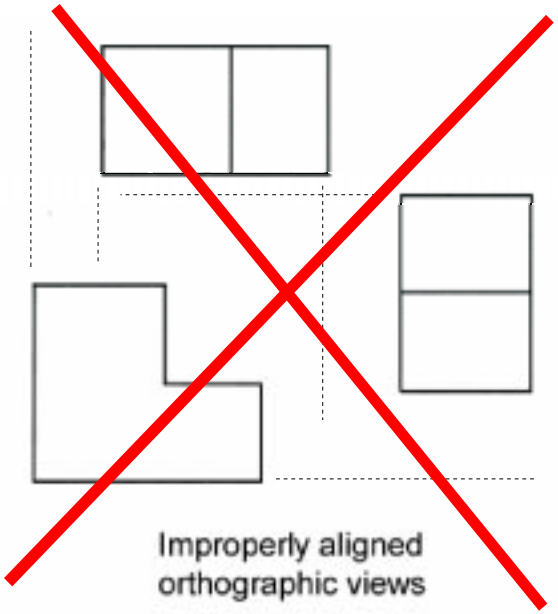


Orthographic Projection

Align views with each other (features project from one view to the next)



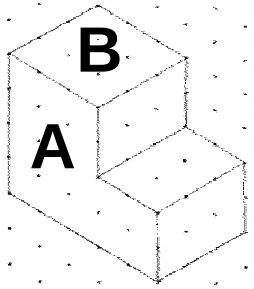
Properly aligned
orthographic views



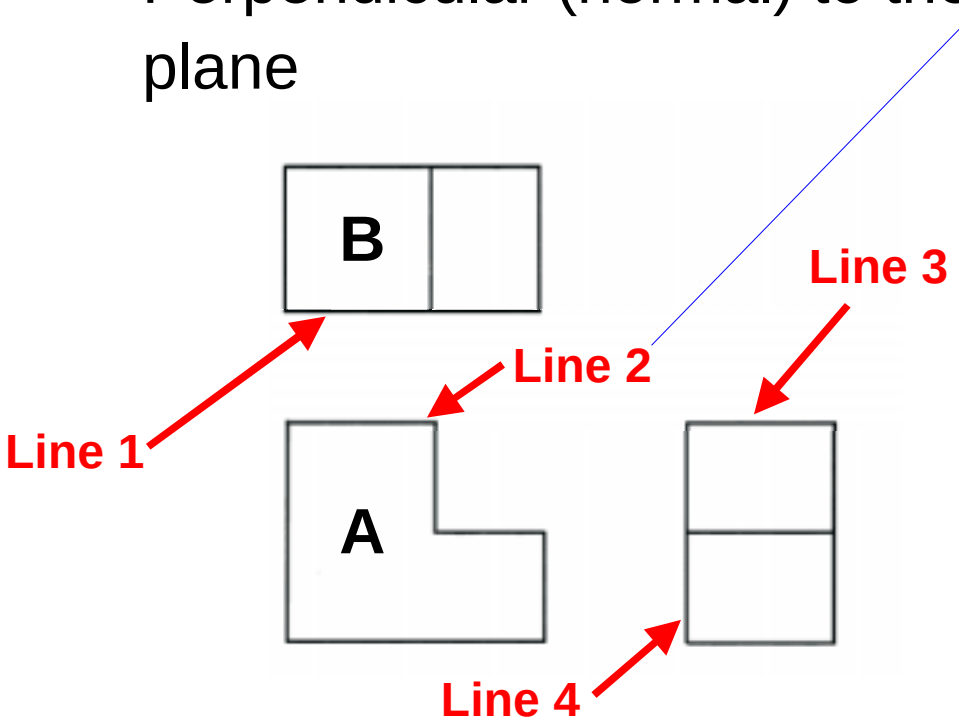
Improperly aligned
orthographic views

Orthographic Projection:

Normal Surfaces



- Parallel to one of the six glass panes of our transparent cube
- Perpendicular (normal) to the projectors to/from that plane



Shown true size and shape in the view that they are parallel to

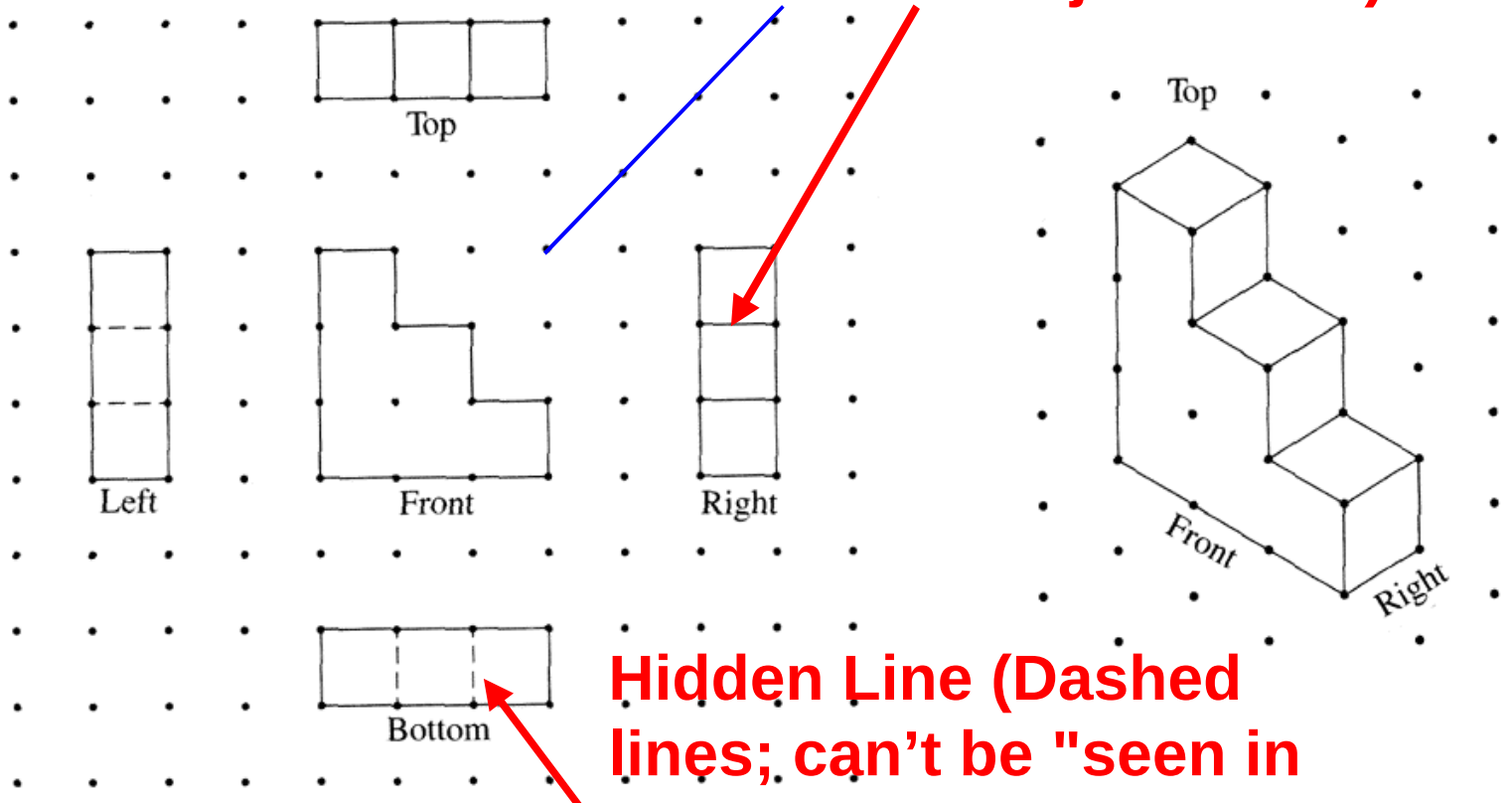
Seen as edges (lines) in the other principal views

A is 1 and 4
B is 2 and 3

Orthographic Projection:

Hidden Lines

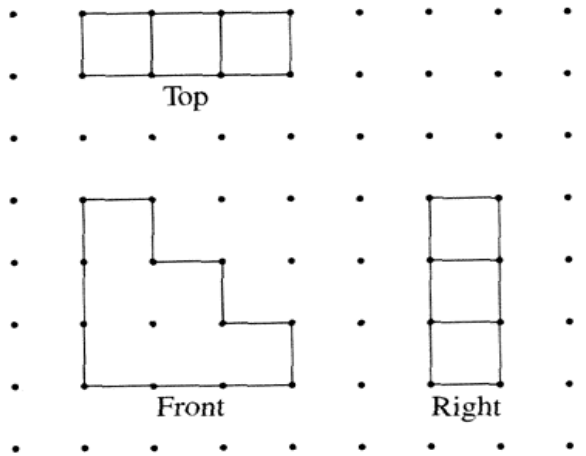
Visible Line (Solid lines, can be seen in a given view, aka: Object Lines)

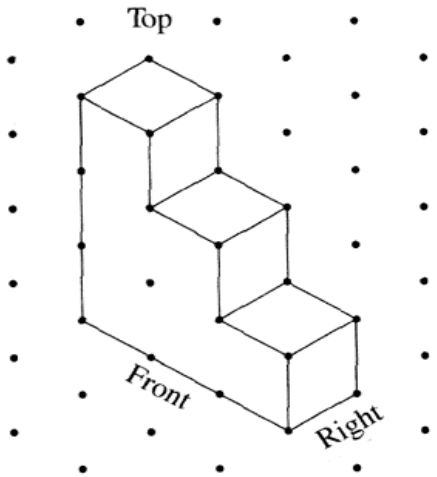


Hidden Line (Dashed lines; can't be "seen in that view)

As an Individual...

- In your lab manual, pp. 28−29 (top two), construct the top, front and right side views of the objects shown in isometric.
  - Make sure your views are aligned and hidden lines are included as necessary
- Compare your answers





Orthographic Projection: Missing Views

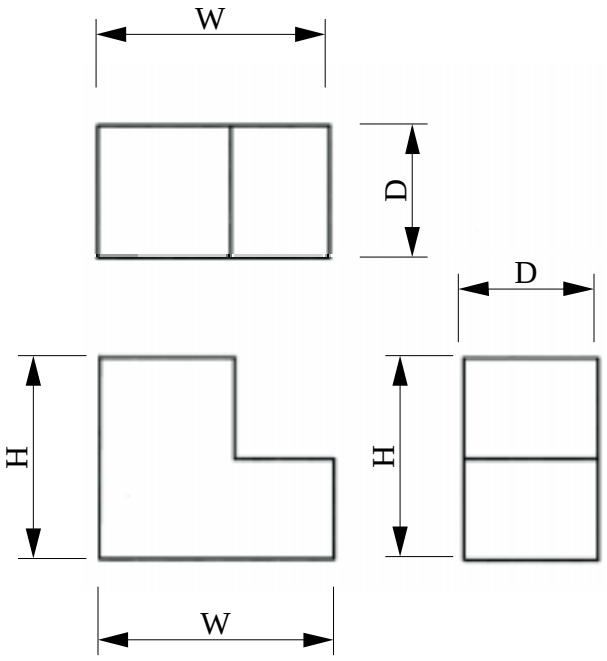
Constructing the third view helps visualization skills

3−D object has edges and surfaces and they appear differently in various views.

- Edge: a point, edge or surface
- Surface: surface or edge

*W*hen constructing the third view, ·ecall:

- Points project perpendicularly from one view to the next
- Object dimensions (height, width and depth) are preserved from one view to the next



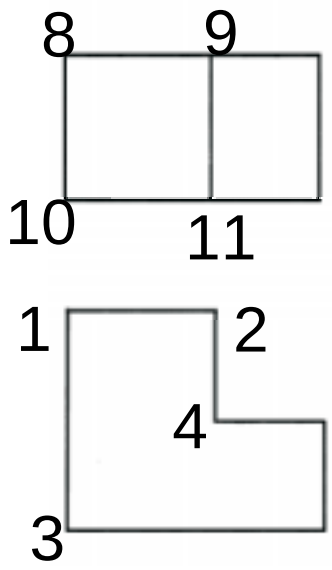
As an Individual...

- In your lab manual, p. 30, construct the third view of the objects
- Compare your answers to those of your teammates

Orthographic Projection:

Surface Identification

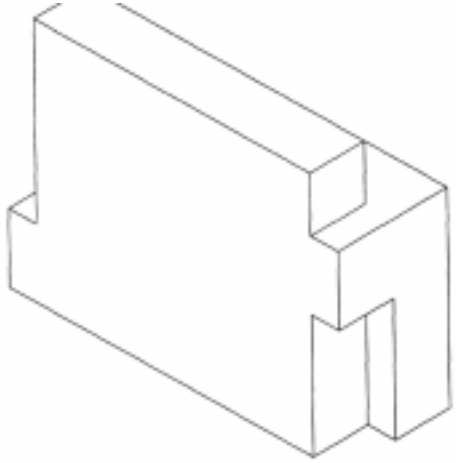
For the indicated surfaces and edges, where are they located in the adjacent view?



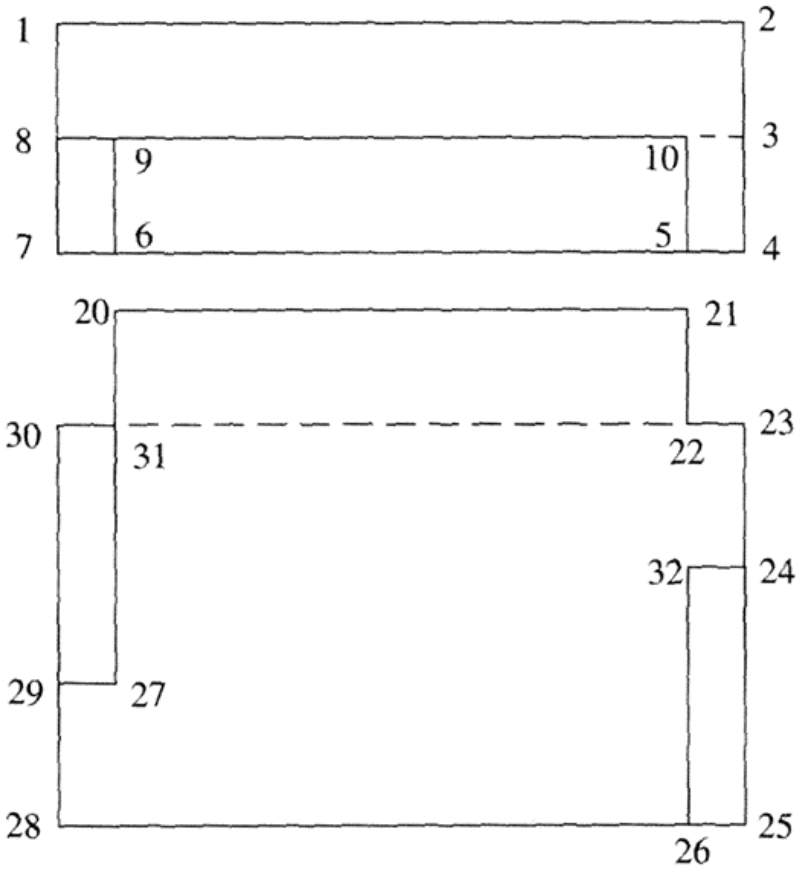
1, 2	Surface	8, 9, 11, 10
3	Edge	10, 8
9,11	Edge	2 or 4

Orthographic Projection:

Surface Identification



For the indicated surfaces and edges,
where are they located in the adjacent view?



30, 23	surface
8, 9	surface
9, 10	surface
10, 3	surface
25	edge
20, 27	edge
29, 27	edge
10, 5	edge

Orthographic Projection:

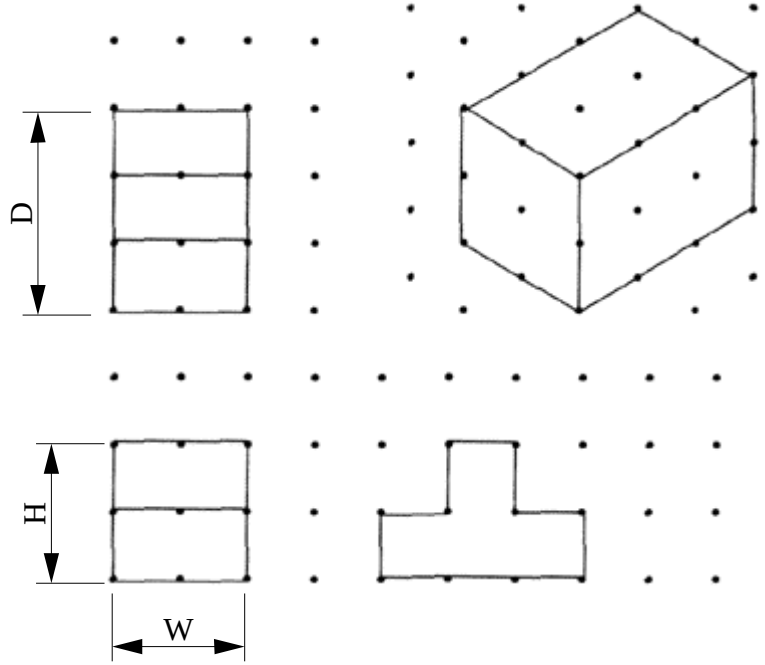
Isometric Sketches from Orthographic Views

Constructing Isometric sketches from Orthographic views develops visualization skills.

Orthographic Projection:

Iso. Sketches from Ortho Sketches

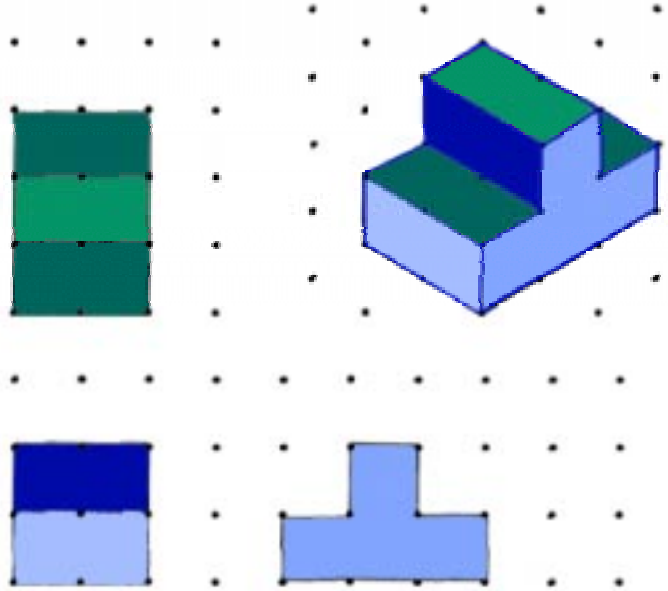
1. Find the object's overall dimensions from the orthographic views and sketch that size box on isometric dot paper.



Orthographic Projection:

Iso. Sketches from Ortho Sketches

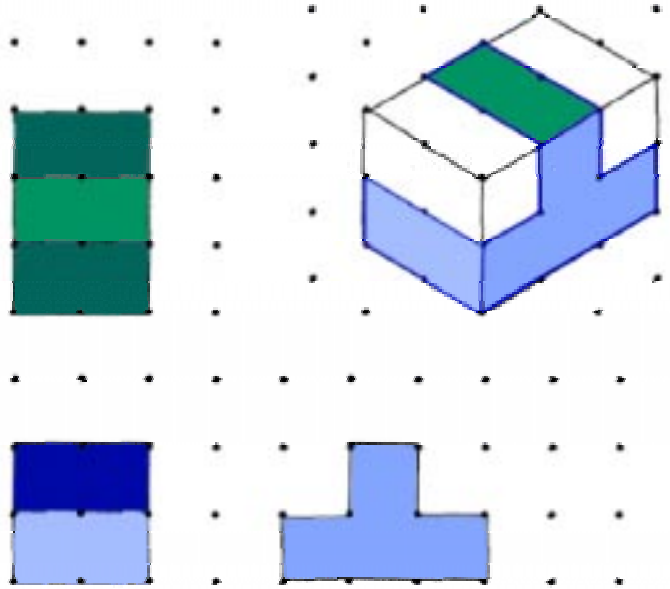
3. Add/remove lines until the view is complete.



Orthographic Projection:

Iso. Sketches from Ortho Sketches

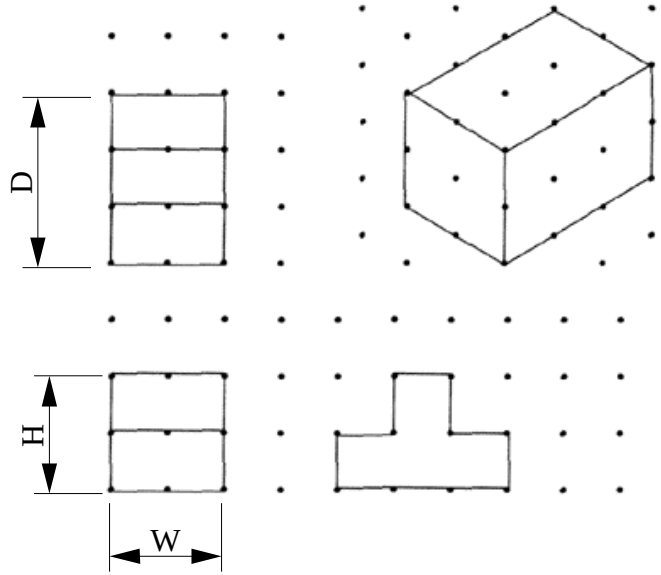
2. Sketch the top, front, and right side views in their appropriate locations on the box.

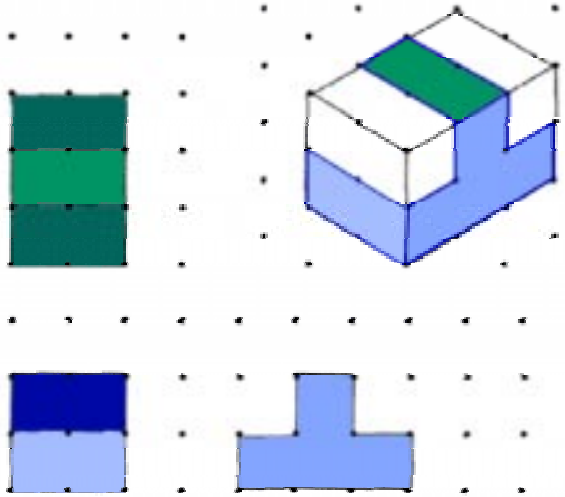


As an Individual...

In your Lab Manual, p. 31:

1. sketch a box of that size on isometric dot paper
2. sketch the top, front, and right side views
3. add and remove lines





Individual Homework

Due: Thursday 8.2

- Complete Lab Manual problems on pp. 28–31 as indicated